

NIKEL, Russian Federation

WMO: 220040

Lat: 69.410N

Lon: 30.209E

Elev: 92

StdP: 100.22

Time Zone: 3.00 (E03)

Period: 87-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.8	-22.1	-28.3	0.3	-25.8	-24.7	0.4	-21.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.2	24.8	15.8	22.1	15.2	20.0	14.4	17.1	23.1	15.7	21.1	14.6	19.4	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.5	10.4	19.8	13.1	9.5	18.0	12.1	8.9	17.2	48.4	23.4	44.3	21.1	41.1	19.8	20.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-29.1	26.9	2.2	2.4	-30.7	28.6	-32.0	30.1	-33.3	31.4	-34.9	33.2		
(5)			-29.3	18.2	2.2	2.0	-30.9	19.6	-32.2	20.7	-33.4	21.9	-35.0	23.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.4	-10.6	-7.6	-4.8	-0.5	5.7	9.6	14.0	12.4	8.6	1.1	-4.3	-7.6
	DBStd	9.44	6.58	5.96	4.89	3.83	4.43	4.48	4.59	3.55	3.58	3.63	4.74	5.73
	HDD10.0	3454	639	492	458	314	147	61	12	13	68	277	428	545
	HDD18.3	6207	898	725	717	564	391	265	149	189	292	536	678	803
	CDD10.0	312	0	0	0	0	15	48	136	87	26	0	0	0
	CDD18.3	23	0	0	0	0	1	3	16	4	0	0	0	0
	CDH23.3	136	0	0	0	0	8	9	101	17	0	0	0	0
	CDH26.7	24	0	0	0	0	1	1	18	4	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	451	33	24	18	16	23	50	66	64	50	37	35	36
	PrecMax	569	65	66	34	39	62	172	145	138	123	66	101	90
	PrecMin	303	2	2	3	3	6	11	13	24	13	9	6	1
	PrecStd	69	14	13	8	8	13	33	34	23	24	14	22	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	4.2	5.2	10.8	23.5	24.3	29.4	25.6	18.8	10.0	4.4	3.6
		MCWB	2.0	2.3	1.7	5.6	14.1	15.7	17.3	17.9	14.0	8.4	2.7	2.0
	2%	DB	2.8	2.4	3.6	7.5	17.8	21.6	26.1	21.9	16.3	8.3	3.2	2.3
		MCWB	1.1	1.1	1.0	4.0	10.2	14.4	16.4	16.1	11.9	6.9	2.0	1.0
	5%	DB	0.0	1.1	2.4	5.3	15.0	19.2	23.7	19.7	14.9	7.0	2.2	1.1
		MCWB	-1.3	-0.2	0.2	2.6	9.0	12.8	15.8	14.8	11.0	5.6	1.3	0.0
	10%	DB	-1.8	-0.2	1.2	3.8	12.4	16.8	21.3	17.7	13.5	5.9	1.4	-0.3
		MCWB	-2.8	-1.3	-0.6	1.6	7.7	11.8	14.6	13.8	10.2	4.7	0.5	-1.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.4	2.6	2.1	6.2	13.6	17.2	19.4	18.9	14.2	8.8	3.2	2.2
		MCDB	3.8	4.0	4.4	10.1	22.1	22.9	23.5	24.4	18.5	9.4	3.9	3.2
	2%	WB	1.0	1.2	1.2	4.2	11.0	15.2	17.6	16.7	12.6	7.0	2.1	1.1
		MCDB	3.0	2.3	3.2	7.2	16.8	20.2	24.0	20.8	15.7	8.4	2.9	2.1
	5%	WB	-1.3	0.0	0.4	2.7	9.2	13.6	16.3	15.2	11.3	5.9	1.2	0.1
		MCDB	-0.1	0.9	2.2	4.6	14.4	18.3	22.9	19.2	14.2	6.9	2.2	1.1
	10%	WB	-2.8	-1.3	-0.4	1.9	7.8	11.7	15.1	13.9	10.4	4.7	0.5	-1.4
		MCDB	-1.8	-0.1	1.2	3.5	12.1	16.2	20.7	17.5	13.1	5.8	1.3	-0.4
Mean Daily Temperature Range	5% DB	MDBR	5.4	4.4	4.6	4.5	5.5	5.5	6.2	4.9	4.1	2.8	3.9	5.2
		MCDBR	5.8	3.8	4.3	4.8	10.1	8.6	9.6	7.6	6.1	2.9	3.6	5.0
		MCWBR	5.3	3.4	3.2	3.1	5.5	4.1	3.8	3.7	3.5	2.7	3.8	4.7
	5% WB	MCDBR	5.7	3.9	4.1	4.4	9.6	8.4	8.3	7.2	5.4	2.9	3.8	4.8
		MCWBR	5.2	3.8	3.4	3.2	5.6	4.6	4.1	4.3	3.6	2.8	3.7	4.6
Clear-Sky Solar Irradiance	taub	0.207	0.253	0.301	0.331	0.356	0.352	0.367	0.349	0.313	0.272	0.361	N/A	
	taud	1.913	2.047	2.124	2.184	2.335	2.443	2.430	2.456	2.516	2.222	2.680	N/A	
	Ebn at Noon	22	475	696	797	826	844	814	786	722	460	57	0	
	Edh at Noon	4	43	80	103	101	94	92	80	57	36	8	0	
All-Sky Solar Radiation	RadAvg	0.01	0.37	1.71	3.42	4.50	4.75	4.39	3.01	1.62	0.48	0.05	0.00	
	RadStd	0.00	0.04	0.09	0.23	0.45	0.38	0.48	0.30	0.13	0.06	0.01	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page